

TRAINING EFFECTIVENESS DASHBOARD

Product Design Document (PDD) + Software Design Document (SDD)

Platform: Microsoft Power BI

Dashboard title: Training Effectiveness Overview

Source: training_effectiveness_data

Purpose

This document defines the business requirements and technical design for the Training Effectiveness Dashboard. It is based on the supplied Power BI report and documents the dashboard's objectives, data model, calculations, visual design, interactions, validation approach and expected business use.

PART A — PRODUCT DESIGN DOCUMENT (PDD)

1. Business Problem

Training teams need a consolidated view of employee participation, completion, learning impact, satisfaction and training spend. Without a centralized analytical view, it is difficult to identify program-level performance, department differences, completion drop-offs and cost efficiency.

2. Product Objective

Provide an interactive Power BI dashboard that enables users to monitor training effectiveness and investigate performance across departments, training programs, months, regions and completion status.

3. Target Users

- HR and Learning & Development teams
- Training managers
- Department managers
- Business/management stakeholders

4. Key Business Questions

- What is the overall training completion rate?
- How much does training improve employee scores?
- What is the average participant satisfaction?
- What is the total training cost?
- How does completion vary by department?
- Which training programs show stronger score improvement?
- How does enrollment vary by month?
- What is the distribution of completed and non-completed training?
- What are the drop-off rate and cost per completion across department/program/region combinations?

5. KPI Requirements

KPI	Purpose
Completion Rate %	Measures the proportion of training participation that reaches completion.
Avg Score Improvement	Tracks the average learning/assessment score improvement.
Avg Satisfaction	Summarizes participant satisfaction with training.
Total Training Cost	Measures overall training expenditure.
Drop-off Rate %	Identifies the proportion of training participation that does not complete.
Cost per Completion	Shows training cost associated with each completion.

6. Dashboard Scope

The supplied report contains one populated report page titled **Training Effectiveness Overview**. It contains KPI cards, a monthly enrollment trend, department completion analysis, program score-improvement analysis, training cost/score-improvement analysis, completion-status distribution, a detailed matrix, and a bookmark navigator.

PART B — SOFTWARE DESIGN DOCUMENT (SDD)

1. Technology Stack

- Microsoft Power BI Desktop
- Power Query for data preparation
- DAX for analytical measures
- Power BI native visuals for reporting
- PBIX as the report/package format

2. Data Source and Main Entity

The report uses the **training_effectiveness_data** table as the primary analytical dataset. A separate **Dax Measures** table is used to organize calculated measures used by the report.

3. Fields Used by the Dashboard

Area	Fields / measures observed in report
Employee / participation	EmployeeID, CompletionStatus, Enrollment Month
Organization	Department, Region
Training	TrainingProgram
Financial	TrainingCost_INR, Total Training Cost, Cost per Completion
Learning outcome	Score Improvement, Avg Score Improvement
Experience	Avg Satisfaction
Effectiveness	Completion Rate %, Drop-off Rate %

4. Visual Design / Report Components

Visual	Implementation / purpose
KPI Card	Displays Completion Rate %, Avg Score Improvement, Avg Satisfaction and Total Training Cost.
Line Chart	Shows count of EmployeeID by Enrollment Month to understand participation trend.
Clustered Bar Chart	Shows Completion Rate % by Department.
Column Chart	Shows Avg Score Improvement by TrainingProgram.
Line & Clustered Column Combo	Combines TrainingCost_INR and score improvement, segmented by TrainingProgram.
Donut Chart	Shows EmployeeID count by CompletionStatus.
Pivot/Matrix Table	Summarizes Department, TrainingProgram and Region with completion, cost, drop-off, cost-per-completion
Bookmark Navigator	Provides report navigation/bookmark interaction support.

5. Measure Design

The report references the following DAX measures: **Completion Rate %**, **Avg Score Improvement**, **Avg Satisfaction**, **Total Training Cost**, **Drop-off Rate %**, and **Cost per Completion**. These measures are designed to remain responsive to report filter context so that the same business metrics can be analyzed at different organizational and training levels.

6. Data Flow

Source data → Power Query / preparation → training_effectiveness_data model → DAX measures → Power BI visuals → interactive dashboard.

7. Interaction Design

Visual selections and report context are intended to cross-filter related analytical views. The matrix provides a detailed breakdown, while charts provide faster pattern discovery. Bookmark navigation supports switching between configured report states where applicable.

8. Validation and Testing

Recommended validation checks include: reconciling EmployeeID counts with source data; validating completion and drop-off calculations against known totals; checking that training costs aggregate correctly; verifying score-improvement calculations; testing department/program/region filter context; checking blank/null handling; and confirming that totals in cards, charts and the matrix remain consistent.

9. Performance Considerations

Use efficient Power Query transformations, avoid unnecessary calculated columns, keep DAX measures context-aware, minimize high-cardinality visuals where possible, and validate model relationships and aggregation behavior before publishing.

10. Deployment

The completed PBIX can be published to Power BI Service for organizational sharing, subject to workspace permissions and the organization's data-refresh/security requirements. If scheduled refresh is required, the underlying source and gateway/connectivity configuration must be established separately.

11. Future Enhancements

- Add slicers for department, region, training program and time period if required.
- Add trainer-level and course-level drill-through pages.
- Add target/benchmark indicators for completion and satisfaction.
- Add cohort or pre/post-training analysis where source data supports it.
- Add automated refresh and Power BI Service deployment documentation.

12. Acceptance Criteria

The dashboard is considered ready when KPI calculations reconcile with the source data, visuals respond correctly to available filters/interactions, the detailed matrix agrees with aggregated visuals, and the report communicates training participation, effectiveness, satisfaction and cost clearly to intended stakeholders.

Document status: Portfolio / academic project documentation based on the supplied Power BI report.